

Inclusion of nonlocality for coated CPA

The system considered in these notes is a coated sphere, where the coating has a spatially nonlocal optical response (the bulk of the sphere is taken to be vacuum or a nondispersive dielectric). The paper [Localized plasmons in graphene-coated nanospheres](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414?casa_token=MyaHrPTqM3QAAAAA%3AEafTsGmUjxm_tjkCxBc-mawHP8qeQWMSdsXgXDYoVT6FeHtW9h9yp8bLtU-7H8kkJnz9vrDKQVsigg) (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414?casa_token=MyaHrPTqM3QAAAAA%3AEafTsGmUjxm_tjkCxBc-mawHP8qeQWMSdsXgXDYoVT6FeHtW9h9yp8bLtU-7H8kkJnz9vrDKQVsigg) is relevant.

Effect of nonlocality on σ_{CPA}

Spatial nonlocality may be introduced by a modification of Ohm's law as follows:

$$\frac{\beta^2}{\omega^2} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega),$$

where β is a quantity that has velocity units and is on the order of the Fermi velocity, v_F . The exact value of β depends on the underlying electronic structure of the coating, so we keep it generic in the following derivation. We note, for convenience, that linear and quadratic dispersions in two dimensions yield the same value of $\beta^2 = 3v_F^2/4$.

In the following, we show that inclusion of spatial nonlocality introduces a modification to σ_{CPA} for TM polarized waves but not for TE polarized waves.

Effect of nonlocality for TM polarized waves

For TM waves, we have that the surface current, $\mathbf{J}(\mathbf{r}, \omega)$, is given as follows:

$$\mathbf{J}(\mathbf{r}, \omega) = \delta(r - R) \left[\nabla \times \mathbf{r} \times \nabla f(\mathbf{r}, \omega) - \mathbf{e}_r \cdot \left[\nabla \times \mathbf{r} \times \nabla f(\mathbf{r}, \omega) \right] \mathbf{e}_r \right],$$

where the Dirac delta function ensures that we only have a current at the location of the coating and we have, of course, ensured that the current is purely tangential to the sphere by subtracting its radial component. In the above, the function $f(\mathbf{r}, \omega) \equiv jY_{lm}(\theta, \phi)$, where j is a constant to be determined later.

For future convenience, we write the current explicitly in spherical coordinates as follows:

$$\mathbf{J}(\mathbf{r}, \omega) = -\frac{\delta(r-R)}{r} \left[\partial_r(r\partial_\theta f(\mathbf{r}, \omega))\mathbf{e}_\theta + \frac{1}{\sin(\theta)}\partial_r(r\partial_\phi f(\mathbf{r}, \omega))\mathbf{e}_\phi \right]$$

At this point, we calculate the divergence of this surface current:

$$\begin{aligned} \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) &= \left[\partial_r \delta(r-R) \right] \mathbf{e}_r \cdot \left[\nabla \times \mathbf{r} \times \nabla f(\mathbf{r}, \omega) \right] - \\ &\frac{1}{r^2} \partial_r \left[r^2 \delta(r-R) \mathbf{e}_r \cdot \left[\nabla \times \mathbf{r} \times \nabla f(\mathbf{r}, \omega) \right] \right] = -\frac{\delta(r-R)}{r^2} \partial_r \left[r^2 \mathbf{e}_r \cdot \left[\nabla \times \mathbf{r} \times \nabla f(\mathbf{r}, \omega) \right] \right] \end{aligned}$$

Note that, quite conveniently, the annoying radial derivatives on the Dirac delta functions cancel out exactly. Therefore, we have:

$$\begin{aligned} \nabla \cdot \mathbf{J}(\mathbf{r}, \omega) &= -\frac{\delta(r-R)}{r^2} \partial_r \left[r^2 \mathbf{e}_r \cdot \left[\nabla \times \left[\partial_\theta f(\mathbf{r}, \omega) \mathbf{e}_\phi - \frac{1}{\sin(\theta)} \partial_\phi f(\mathbf{r}, \omega) \mathbf{e}_\theta \right] \right] \right] = \\ &-\frac{\delta(r-R)}{r^2} \partial_r \left[\frac{r}{\sin(\theta)} \left[\partial_\theta \left(\sin(\theta) \partial_\theta f(\mathbf{r}, \omega) \right) + \frac{1}{\sin(\theta)} \partial_\phi^2 f(\mathbf{r}, \omega) \right] \right] = \\ &\frac{\delta(r-R)l(l+1)}{r^2} \partial_r \left[r f(\mathbf{r}, \omega) \right], \end{aligned}$$

where we used the fact that the spherical harmonics, $Y_{lm}(\theta, \phi)$, are eigenfunctions of the square of the angular momentum operator.

To complete our derivation, we now calculate the gradient of the divergence obtained above:

$$\nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) = \frac{\delta(r-R)l(l+1)}{r^3} \left[\partial_r \left[r \partial_\theta f(\mathbf{r}, \omega) \right] \mathbf{e}_\theta + \frac{1}{\sin(\theta)} \partial_r \left[r \partial_\phi f(\mathbf{r}, \omega) \right] \mathbf{e}_\phi \right]$$

Therefore, we have:

$$\begin{aligned} \frac{\beta^2}{\omega^2} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \mathbf{J}(\mathbf{r}, \omega) &= \left[1 - \frac{\beta^2 l(l+1)}{\omega^2 R^2} \right] \mathbf{J}(\mathbf{r}, \omega) \rightarrow \mathbf{J}(\mathbf{r}, \omega) = \\ &\frac{\sigma(\omega)}{1 - \beta^2 l(l+1)/(R^2 \omega^2)} \mathbf{E}(\mathbf{r}, \omega) \end{aligned}$$

Thus, we immediately have the following nonlocal correction to σ_{CPA} :

$$\sigma_{CPA}^{NL}(k^+) = \left[1 - \frac{(\beta/c)^2 l(l+1)}{(k^+)^2} \right] \sigma_{CPA}(k^+),$$

where $k^+ \equiv R\omega/c$.

Effect of nonlocality for TE polarized waves

For TE polarized waves, there is no correction due to nonlocality. This follows simply from the fact that, in this case, the surface current has the form:

$$\mathbf{J}(\mathbf{r}, \omega) = \delta(r - R) \nabla \times (\mathbf{r} f(\mathbf{r}, \omega)),$$

which has no divergence. Note that in the above, we did not have to subtract any radial component since the angular momentum operator has no radial component.

Cylinders

We now turn our attention to cylinders. In this case, the current is given by (with $f(\mathbf{r}, \omega) = j\rho$, with j being a constant—note that f requires a ρ dependence due to the derivatives with respect to ρ below):

$$\mathbf{J}(\mathbf{r}, \omega) = \delta(r - \rho) \left[\nabla \times (f(\mathbf{r}, \omega) e^{im\theta} \mathbf{e}_z) - \mathbf{e}_\rho \cdot \left[\nabla \times (f(\mathbf{r}, \omega) e^{im\theta} \mathbf{e}_z) \right] \right],$$

where, just as in the last section, we ensured a null normal component of the surface current. We write the current explicitly in cylindrical coordinates as follows:

$$\mathbf{J}(\mathbf{r}, \omega) = -\delta(\rho - R) e^{im\theta} \partial_\rho f(\mathbf{r}, \omega) \mathbf{e}_\phi$$

We now turn to the relevant derivatives of the current:

$$\nabla \cdot \mathbf{J}(\mathbf{r}, \omega) = -\frac{im}{R} \delta(\rho - R) e^{im\theta} \partial_\rho f(\mathbf{r}, \omega), \quad \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) = \frac{m^2}{R^2} \delta(\rho - R) e^{im\theta} \partial_\rho f(\mathbf{r}, \omega)$$

Therefore, we obtain:

$$\frac{\beta^2}{\omega^2} \nabla(\nabla \cdot \mathbf{J}(\mathbf{r}, \omega)) + \mathbf{J}(\mathbf{r}, \omega) = \left[1 - \frac{\beta^2 m^2}{\omega^2 R^2} \right] \mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega)$$

Therefore, we obtain the following nonlocal correction to the CPA conductivity:

$$\sigma_{CPA}^{NL}(k^+) = \left[1 - \frac{(\beta/c)^2 m^2}{(k^+)^2} \right] \sigma_{CPA}(k^+),$$